Algebra 1

100 Practice Problems

Organized by topic • Increasing difficulty

How this set is ordered. The eleven topics run in the order a standard Algebra 1 course teaches them, so the set gets harder from front to back. Within each topic the problems also climb: the first one or two are direct practice, the last one or two combine the topic with something earlier. Problems 1–16 need only arithmetic; by problems 93–100 you are factoring, applying the quadratic formula, and reading a vertex out of a word problem.

Answer key begins on the last pages. Answers are exact — fractions and radicals are left in exact form rather than rounded, except where a problem asks for a rounded value.

1. Variables, Expressions & Order of Operations

Evaluate. Follow the order of operations; watch signs when substituting negative numbers.

1. Evaluate $3 + 4 \times 5$.

2. Evaluate $2(7 - 3)^2$.

3. Evaluate $5x - 2$ when $x = 4$.

4. Evaluate $x^2 + 3x - 1$ when $x = -2$.

5. Evaluate $\dfrac{a + b}{a - b}$ when $a = 7$ and $b = 3$.

6. Evaluate $-3(2x - 5) + 4x$ when $x = -1$.

7. Evaluate $2x^2y - xy^2$ when $x = 3$ and $y = -2$.

8. Evaluate $\dfrac{4 + 2(3^2 - 7)}{-2}$.

2. Properties of Real Numbers & Simplifying Expressions

Simplify completely. Distribute before combining like terms, and treat a leading minus sign as -1.

9. Simplify $7x + 3x - 2x$.

10. Simplify $4(x + 3)$.

11. Simplify $5(2x - 1) - 3x$.

12. Simplify $-2(3a - 4) + 5(a + 2)$.

13. Simplify $3(2x - 5) - 2(4x - 7)$.

14. Simplify $\frac{1}{2}(6x - 8) + \frac{2}{3}(9x + 3)$.

15. Simplify $4x - \big[\,3 - 2(x - 5)\,\big]$.

16. Simplify $-\big[\,2(3x - 1) - (x + 4)\,\big]$.

3. One-Variable Linear Equations

Solve for the variable. Problems 25–28 involve fractions, decimals, or solving for one variable in terms of another.

17. Solve $x + 7 = 12$.

18. Solve $3x = -21$.

19. Solve $2x - 5 = 11$.

20. Solve $\frac{x}{4} + 3 = 7$.

21. Solve $5x + 2 = 3x + 14$.

22. Solve $4(x - 3) = 20$.

23. Solve $3(2x + 1) = 4x + 15$.

24. Solve $2(x - 4) + 5 = 3(x + 1) - 6$.

25. Solve $\frac{2x - 1}{3} = \frac{x + 4}{2}$.

26. Solve $\frac{x}{2} - \frac{x}{5} = 3$.

27. Solve $3x + 2y = 12$ for y.

28. Solve $0.05x + 0.10(x + 3) = 1.50$.

4. Linear Inequalities

Solve and write the solution as an inequality. Remember to reverse the symbol when you multiply or divide by a negative number.

29. Solve $x + 5 < 9$.

30. Solve $3x \geq -12$.

31. Solve $-2x > 10$.

32. Solve $4x - 7 \leq 9$.

33. Solve $5 - 3x > 17$.

34. Solve $2(x + 3) \leq 4x - 2$.

35. Solve the compound inequality $-3 < 2x + 1 \leq 7$.

36. Solve $|x - 3| \leq 5$.

5. Ratios, Proportions & Percents

Set up a proportion or a percent equation before solving. The last two are multi-step applications.

37. Solve the proportion $\frac{x}{6} = \frac{10}{15}$.

38. Solve the proportion $\frac{3}{x} = \frac{12}{20}$.

39. What is 15% of 80?

40. 18 is what percent of 45?

41. A shirt costs \$40 and is marked down 25%. What is the sale price?

42. On a map, 2 cm represents 15 km. How many kilometers does 7 cm represent?

43. A town's population grows from 250 to 310. What is the percent increase?

44. How many liters of a 20% acid solution must be added to 10 liters of a 50% acid solution to produce a 30% solution?

6. Functions & Relations

Use function notation carefully: $f(-3)$ means substitute -3 for every x. Problems 50–52 ask about domain and about substituting an expression.

45. Is the relation $\{(1,2), (2,4), (3,6)\}$ a function? Explain in one sentence.

46. Is the relation $\{(1,2), (1,3), (2,5)\}$ a function? Explain in one sentence.

47. If $f(x) = 3x - 4$, find $f(5)$.

48. If $f(x) = x^2 - 2x$, find $f(-3)$.

49. If $f(x) = 2x + 1$, find the value of x for which $f(x) = 11$.

50. State the domain of $f(x) = \sqrt{x - 4}$.

51. State the domain of $f(x) = \dfrac{x + 2}{x - 3}$.

52. If $g(x) = x^2 - 5$, find and simplify $g(a + 1)$.

7. Linear Functions: Slope & Graphing

Slope is $m = \dfrac{y_2 - y_1}{x_2 - x_1}$. Give every equation in slope-intercept form $y = mx + b$ unless told otherwise.

53. Find the slope of the line through $(0, 0)$ and $(2, 6)$.

54. Find the slope of the line through $(1, 3)$ and $(5, 11)$.

55. Find the slope of the line through $(-2, 5)$ and $(4, -7)$.

56. State the slope of $y = -3x + 7$.

57. State the y-intercept of $2x + y = 8$.

58. Write the equation of the line with slope 4 passing through $(0, -3)$.

59. Write the equation of the line through $(2, 5)$ with slope 3.

60. Write the equation of the line through $(1, 4)$ and $(3, 10)$.

61. Rewrite $3x - 4y = 12$ in slope-intercept form.

62. What is the slope of any line parallel to $y = \frac{2}{5}x + 1$?

63. What is the slope of any line perpendicular to $y = -\frac{3}{4}x + 2$?

64. Write the equation of the line through $(-1, 2)$ that is perpendicular to $y = 2x - 5$.

8. Systems of Linear Equations

Solve by substitution or elimination. Two of these have no single solution — say which case it is and why.

65. Solve: $y = 2x$ and $x + y = 9$.

66. Solve: $y = x + 1$ and $y = 3x - 5$.

67. Solve: $x + y = 10$ and $x - y = 4$.

68. Solve: $2x + y = 7$ and $3x - y = 8$.

69. Solve: $3x + 2y = 16$ and $5x - 2y = 8$.

70. Solve: $4x + 3y = 10$ and $2x - y = 0$.

71. Solve: $2x + 3y = 12$ and $4x + 6y = 24$.

72. Solve: $x - 2y = 5$ and $2x - 4y = 3$.

73. Two numbers have a sum of 30 and a difference of 8. Find the numbers.

74. Three adult tickets and two child tickets cost \$34. Two adult tickets and five child tickets cost \$41. Find the price of each ticket.

9. Exponents & Exponential Functions

Write answers with positive exponents only. The last two problems build an exponential model from a description.

75. Simplify $x^3 \cdot x^5$.

76. Simplify $\left(x^4\right)^3$.

77. Simplify $\dfrac{x^9}{x^4}$.

78. Simplify $\left(2x^3y\right)^4$.

79. Simplify $5x^0 + 3^{-2}$.

80. Simplify $\dfrac{3x^2y^{-3}}{6x^{-1}y^2}$.

81. A colony of 500 bacteria doubles every year. Write a function $P(t)$ for the population after t years, then find $P(3)$.

82. A car worth \$24,000 loses 15% of its value each year. Write a function $V(t)$ for its value after t years, then find $V(3)$.

10. Polynomials & Factoring

Add, subtract, and multiply first; then factor completely. Problem 92 has a leading coefficient other than 1.

83. Add: $\left(3x^2 + 2x - 1\right) + \left(x^2 - 5x + 4\right)$.

84. Subtract: $\left(5x^2 - x + 2\right) - \left(2x^2 + 3x - 6\right)$.

85. Multiply: $3x\left(2x^2 - 4x + 1\right)$.

86. Multiply: $(x + 3)(x - 5)$.

87. Multiply: $(2x - 3)(3x + 4)$.

88. Factor completely: $6x^2 + 9x$.

89. Factor completely: $x^2 - 49$.

90. Factor completely: $x^2 + 7x + 12$.

91. Factor completely: $x^2 - 5x - 24$.

92. Factor completely: $6x^2 + 11x - 10$.

11. Quadratic Equations & Functions

Solve by square roots, factoring, or the quadratic formula $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The last two ask for a vertex.

93. Solve $x^2 = 36$.

94. Solve $x^2 - 9 = 0$.

95. Solve $x^2 + 5x + 6 = 0$.

96. Solve $x^2 - 7x + 12 = 0$.

97. Solve $2x^2 - 5x - 3 = 0$.

98. Solve $x^2 + 4x - 1 = 0$. Leave your answer in exact form.

99. Find the vertex of $f(x) = x^2 - 6x + 5$.

100. A ball is thrown upward and its height in feet after t seconds is $h(t) = -16t^2 + 64t$. Find the maximum height and the time it is reached.

Answer Key

Expressions (1–8)

1. 23 **2.** 32 **3.** 18
4. -3 **5.** $\frac{5}{2}$ **6.** 17
7. -48 **8.** -4

Simplifying (9–16)

9. $8x$ **10.** $4x + 12$
11. $7x - 5$ **12.** $-a + 18$
13. $-2x - 1$ **14.** $9x - 2$
15. $6x - 13$ **16.** $-5x + 6$

Linear Equations (17–28)

17. $x = 5$ **18.** $x = -7$
19. $x = 8$ **20.** $x = 16$
21. $x = 6$ **22.** $x = 8$
23. $x = 6$ **24.** $x = 0$
25. $x = 14$ **26.** $x = 10$
27. $y = 6 - \frac{3}{2}x$ **28.** $x = 8$

Inequalities (29–36)

29. $x < 4$ **30.** $x \geq -4$
31. $x < -5$ **32.** $x \leq 4$
33. $x < -4$ **34.** $x \geq 4$
35. $-2 < x \leq 3$ **36.** $-2 \leq x \leq 8$

Ratios & Percents (37–44)

37. $x = 4$ **38.** $x = 5$
39. 12 **40.** 40%
41. \$30 **42.** 52.5 km
43. 24% increase
44. 20 liters

Functions (45–52)

45. Yes — each input has exactly one output.
46. No — the input 1 pairs with both 2 and 3.
47. 11 **48.** 15 **49.** $x = 5$
50. $x \geq 4$
51. All real numbers except $x = 3$
52. $a^2 + 2a - 4$

Slope & Lines (53–64)

53. 3 **54.** 2 **55.** -2
56. -3 **57.** $(0, 8)$
58. $y = 4x - 3$ **59.** $y = 3x - 1$
60. $y = 3x + 1$
61. $y = \frac{3}{4}x - 3$
62. $\frac{2}{5}$ **63.** $\frac{4}{3}$
64. $y = -\frac{1}{2}x + \frac{3}{2}$

Systems (65–74)

65. $(3, 6)$ **66.** $(3, 4)$
67. $(7, 3)$ **68.** $(3, 1)$
69. $\left(3, \frac{7}{2}\right)$ **70.** $(1, 2)$
71. Infinitely many solutions — the second equation is twice the first, so both describe the same line.
72. No solution — the lines have equal slopes but different intercepts, so they are parallel.
73. 19 and 11
74. Adult \$8, child \$5

Exponents (75–82)

75. x^8 **76.** x^{12} **77.** x^5
78. $16x^{12}y^4$ **79.** $\frac{46}{9}$
80. $\dfrac{x^3}{2y^5}$
81. $P(t) = 500 \cdot 2^t$; $P(3) = 4000$
82. $V(t) = 24000(0.85)^t$; $V(3) = \$14{,}739$

Polynomials (83–92)

83. $4x^2 - 3x + 3$
84. $3x^2 - 4x + 8$
85. $6x^3 - 12x^2 + 3x$
86. $x^2 - 2x - 15$
87. $6x^2 - x - 12$
88. $3x(2x + 3)$ **89.** $(x + 7)(x - 7)$
90. $(x + 3)(x + 4)$
91. $(x - 8)(x + 3)$
92. $(3x - 2)(2x + 5)$

Quadratics (93–100)

93. $x = \pm 6$ **94.** $x = \pm 3$
95. $x = -2,\ -3$ **96.** $x = 3,\ 4$
97. $x = -\frac{1}{2},\ 3$
98. $x = -2 \pm \sqrt{5}$
99. $(3, -4)$
100. 64 feet, reached at $t = 2$ seconds